

Tutorial 8 – Redesign the NUS NextBus System (Take-home tutorial)

Objective

Apply concepts from L10 "Building Information Systems" to analyze and redesign the NUS NextBus system.

Duration

1 – 1.5 hours

Activity Instructions

1. Download and familiarize yourself with the NUS “NextBus” app.
2. Answer the following questions:
 - a. Analyze the Business Process Management (BPM) of the NextBus application.

Aspect	Guiding Questions
Process Identification	What are the key business processes supported by the NextBus app?
Process Analysis	Are these processes smooth and efficient? Are there issues like delays, inaccurate information, or poor user experience?
Process Redesign	Are there any areas for improvement?
Implementation and Monitoring	Does NextBus continuously collect user feedback and update the system? Is there monitoring of performance metrics (e.g., accuracy of arrival time predictions)?

- b. What system conversion strategy did NUS use to deploy the NextBus app, and why do you think this strategy was chosen?

- Identify the specific conversion strategy (e.g., parallel, direct cutover, pilot, phased).
- Explain the potential benefits and risks of the selected approach in the NUS campus context.

- c. Draw a system structure chart and a data flow diagram (DFD) for the current NextBus system.

- Show the major modules and functions in the structure chart.
- Illustrate the flow of data between users, databases, and system components in the DFD.

d. Redesign the “Directions Search” module for NextBus to enable real-time, step-by-step campus navigation for international students and guests, dynamically guiding them from their dormitory to their lecture venue using both shuttle services and walking routes, adjusting based on live bus arrivals and users’ real-time locations.

Briefly describe how you would carry out the six steps of the System Development Process:

- 1) Systems Analysis: What problems will you identify, and what information requirements will you specify?
- 2) Systems Design: What key design choices (interface, input, output) will you make?
- 3) Programming: How will you approach developing the new module?
- 4) Testing: How will you test the functionality to ensure it works correctly and reliably?
- 5) Conversion: How will you implement the new module without disrupting existing services?
- 6) Production and Maintenance: How will you monitor the system post-deployment and plan for updates or improvements?

3. Individual Submission (On Canvas as per the deadline indicated for Tutorial 8)

- a. Each student must individually submit answers to the questions in a PDF file of no more than two pages, named: T8_<full-name>.pdf(e.g., T8_xiaoming.pdf).
- b. Students must submit their work by the deadline to receive score for this tutorial.
- c. Delayed submissions will NOT get any score.